Table 1: **IoT Platforms Focused on Connectivity**

Details	Cisco IoT Control Center	EMnify	Vodafone IoT
Type	SaaS solution	SaaS solution	SaaS solution
Account Types	2 - Essential (Simple IoT) and Advantage (Complex IoT). Pricing, features, and services differ in each type.	EMnify has a single plan type but has two types of coverage options -- Basic and Extended	N/A
API Support	Available	Available	Available
Visualisation	The Control Centre is an environment for real-time evaluation, testing, and deployment	Dashboard	Dashboard, Web interface
Services	Data (GPRS, LTE, NB-IoT, 5G NSA, 5G SA), SMS, Voice	Data (2G, 3G, 4G, LTE-M, NB-IoT), SMS, USSD	LTE, LPWA Technologies, 4G, 5G, NB-IoT, SMS, Voice
Pricing	Multiple pricing plans are available - monthly fixed pool, monthly flexible pool, prepaid and monthly individual. Pricing depends on the account type	Pricing varies based on the number of active SIMS and device consumption. They have a pay-as-you-go option as well as an inclusive volume option	Outcome-based pricing depending on your solution.
Additional Features	Multiple add-on packages that enterprises can subscribe to as their needs evolve	Automated integration into cloud (AWS, Azure, Google Cloud)	Vodafone Business App-Invent: This is an Application-as-a-Service solution that helps companies create IoT apps quickly and cost-effectively.
Getting Started	Cisco provides a Starter Kit, making it easier to launch your connected device services	The company provides a free trial as well as a Factory Test Mode for pre-deployment connectivity testing.	N/A (Contact Vodafone IoT to obtain onboarding details)
Website	*https://www.cisco.com/c/en/us/solutions/internet-of-things/iot-control-center.html*	*https://www.emnify.com/*	*https://www.vodafone.com/business/iot/managed-iot-connectivity/iot-platform*

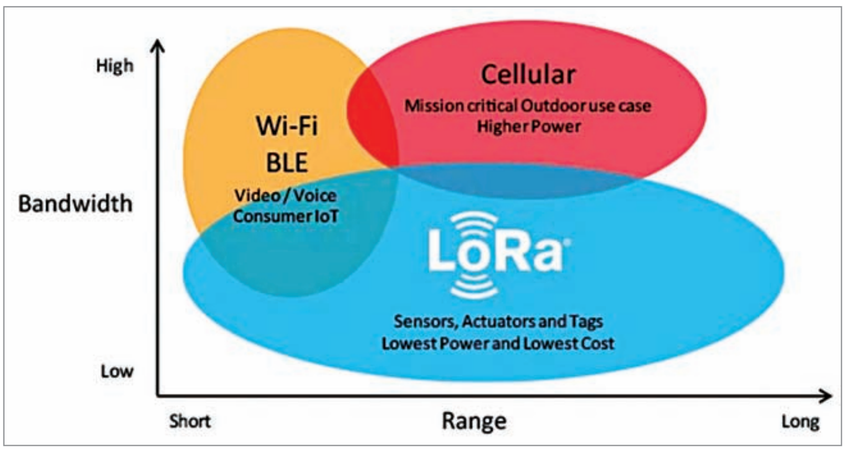
4. Customer support

Another important parameter to consider is whether your platform comes with the necessary support for your application. Also, if you do not have cloud engineers in your team, you can consider a vendor that provides in-person support in setting up your platform. Certain platforms cater to only software and cloud needs. In such a scenario, if your application is more hardware-centric, you will end up doing a lot of work yourself.

5. User-friendliness

A complex platform that is not easy to use involves a learning curve even for IoT professionals. But it does not make sense to spend time learning the platform instead of working on the solution itself. As a decision-maker, you need to consider the user-friendliness of the platform—whether the platform offers dashboards, graphs, and reports.

Young IoT platforms may not have user-friendly user interface (UI) as the developers are more concerned with connectivity, data security, and other issues. Before selecting, you can try using the trial version of the platforms you shortlisted, just to check if these are flexible and user-friendly.



Bandwidth vs range of connection protocols (Credit: The Things Network)

6. Developer-friendliness

Does the platform connect easily with your existing systems, including the IoT hardware? Does it provide a good test and development environment? Does it use open source technologies that your developers might already be aware of? Perhaps the most critical feature of any IoT platform is how easy it is for developers to use.

You need to ensure that the platform supports integrations like RTOS and SDKs in various programming languages, and API support. Many platforms provide custom APIs during the implementation phase, which makes it easy for developers to get started.

7. Disaster recovery and downtime

In any application, it is only natural for technical problems to occur.